

$$A \rightarrow A/I_1 \times A/I_2$$

① If $I_1 + I_2 = A$, then $I_1 I_2 = I_1 \cap I_2$

② φ is surjective iff $I_1 + I_2 = A$

③ φ is injective iff $I_1 \cap I_2 = \{0\}$.

Let $I_1, \dots, I_n$ be ideals

$$\varphi: A \rightarrow A/I_1 \times A/I_2 \times \dots \times A/I_n$$

① $\forall i \neq j, I_i + I_j = A \Rightarrow \prod_{i=1}^n I_i = \bigcap_{i=1}^n I_i$ } $\textcircled{\text{Ex}}$

② φ is surjective iff $I_i \& I_j$ are co-prime $\forall i \neq j$

③ φ is injective $\bigcap_{i=1}^n I_i = \{0\}$.

$$I \subseteq \mathcal{P}_1 \cup \mathcal{P}_2 \Rightarrow I \subseteq \mathcal{P}_1 \text{ or } I \subseteq \mathcal{P}_2?$$

Let $x \in I$ & Assume $I \not\subseteq \mathcal{P}_1$ i.e. $\exists y \in I \setminus \mathcal{P}_1$

$\Rightarrow y \in \mathcal{P}_2, x+y \in I$, suppose $x+y \in \mathcal{P}_1$

If $x \in \mathcal{P}_1$, then $y \in \mathcal{P}_1$ $\Rightarrow$ $\textcircled{\text{}} \Leftarrow$

$\Rightarrow x \notin \mathcal{P}_1 \Rightarrow x \in \mathcal{P}_2$

$\Rightarrow I \subseteq \mathcal{P}_2$

